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Construction and Evaluation of a Large Language Model for the Energy Domain Leveraging Retrieval-Augmented Generation

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***Abstract*—Knowledge in the energy sector is characterized by its multi-source heterogeneity, strong specialization, and rapid evolution, posing significant challenges to traditional information retrieval and question-answering systems. To address this, this study constructs a regulatory Q&A large language model. By adopting Retrieval-Augmented Generation technology, we collected and built an energy knowledge base, constructed a retriever and a generator, and implemented a vertical-domain LLM Q&A system. An experimental dataset was created for evaluation, resulting in scores of 0.93 for answer similarity, 0.79 for answer relevance, and 0.63 for answer correctness. The proposed model outperformed the base model by 0.04, 0.02, and 0.05 points, respectively. It effectively mitigates issues of the base model in vertical domains, such as inadequate semantic understanding, unhelpful responses, and factual hallucinations, thereby providing an effective technical pathway for intelligent knowledge services in the energy field.**

Keywords—Retrieval-Augmented Generation (RAG); Large Language Model (LLM); Energy Knowledge Base; Hallucination

I. Introduction

Energy serves as the cornerstone of national economic and social development. Driven by the dual forces of energy transition and digital transformation, fields such as electric power, carbon neutrality, and renewable energy are generating massive amounts of unstructured data, including national policies, industry standards, technical patents, scientific research papers, and market analysis reports. How to rapidly and accurately acquire knowledge from these multi-source, heterogeneous datasets and transform it into decision support has become a pressing need for the industry's development.

In recent years, generative artificial intelligence, particularly large language models (LLMs), has made remarkable breakthroughs, demonstrating impressive capabilities in general-domain dialogue and content generation. However, their direct application to highly specialized vertical domains like energy has revealed inherent limitations: poor knowledge timeliness, as model training data has a cutoff date, hindering access to the latest policy and technical updates; insufficient domain knowledge, due to the low density of specialized content in pre-training corpora, leading to superficial or erroneous understanding of complex concepts; and the "hallucination" problem, where models generate seemingly plausible but factually incorrect professional content, which is critically unacceptable in the accuracy-sensitive energy sector.

To address these challenges, this paper explores and implements the advanced technical pathway of Retrieval-Augmented Generation (RAG). RAG integrates parametric LLMs with non-parametric external knowledge bases, effectively combining the model's powerful generative capacity with the accuracy and timeliness of the knowledge base. The main contributions of this paper include: designing and implementing a specialized construction and processing pipeline for an energy-domain knowledge base; and building a large language model for energy knowledge base question-answering by leveraging RAG technology and a parameter-efficient fine-tuning framework for LLMs.

II. Related Work

A. Development of question answering system

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Molla et al. [1] define a question-answering system as one capable of responding to arbitrary questions posed in natural language. The earliest QA systems operated on a retrieval-based principle, identifying answers by matching user-provided keywords against a dataset. A notable example is ELIZA [2], which could engage in interactive dialogues with users. By 2018, the advent of deep learning–based large language models, exemplified by BERT [3], marked a significant evolution. These models demonstrated the ability to comprehend and generate human-like natural language, thereby enabling QA systems to deliver more accurate and contextually appropriate responses. In late 2022, ChatGPT [4], built on a decoder-only architecture and trained on massive unlabeled corpora, further advanced the field by acquiring generalized knowledge and linguistic patterns. This breakthrough endowed QA systems with remarkable natural language understanding and sophisticated content generation capabilities.

B. Development of large language models

A large language model (LLM) is a pre-trained linguistic model built upon massive text data. By learning extensive general knowledge and patterns, it demonstrates strong generalization capabilities in processing human language. Based on the Transformer architecture, LLMs can be categorized into three types [5]: the Encoder-Only architecture (e.g., BERT, RoBERTa, ALBERT), the Encoder-Decoder architecture (also known as sequence-to-sequence architecture, exemplified by T5), and the Decoder-Only architecture (referred to as generative architecture, including the GPT series, LLaMA series, and Qwen series) [6]. The Decoder-Only models are particularly notable for their capacity for unsupervised pre-training, enabling them to acquire general knowledge and patterns from large-scale unlabeled corpora.

C. The deficiencies of large language models in specific domains

A prominent challenge in current large language models (LLMs) is the phenomenon of hallucination. LLM hallucination refers to the generation of grammatically correct, fluent, and seemingly authentic responses that, however, contradict input information, deviate from factual accuracy, or present unsubstantiated content [7]. These hallucinations are generally categorized into input conflict, context conflict, and factual conflict [8].

Factual conflict is of particular concern in vertical applications. The parameters of LLMs implicitly store prior knowledge acquired during pre-training. However, due to variations in the quality and scope of pre-training corpora, the knowledge integrated into LLMs may be incomplete or outdated. Consequently, LLMs often underperform when directly applied to specialized vertical domains, primarily due to limitations in knowledge boundaries, outdated knowledge, and inherent knowledge biases.

III. TECHNICAL OPTIMIZATION

This chapter presents the key contributions of this research: the database design philosophy and the retriever optimization.

A. Building a Vertical-Specific Knowledge Base

A vertical domain knowledge base is a system designed to meet precise, professional information needs within a specific industry, as opposed to general-purpose knowledge systems. In the energy sector, this translates to addressing the deep information requirements of industry practitioners, researchers, and decision-makers regarding technical specifications, policies and regulations, project cases, and operational data related to clean energy sources such as hydropower and photovoltaics. Therefore, the construction of a hydropower and photovoltaic energy knowledge base must be closely aligned with the professional characteristics of the field and the core demands of its users, ensuring accuracy, timeliness, and ease of application.

B. Data Collection

Data collection serves as the foundational step in constructing the knowledge base. To ensure its comprehensiveness and authority, the research team systematically aggregated data related to hydropower and photovoltaics from multiple sources, accumulating over 1,200 core documents and more than 500,000 structured data points. The collected materials cover the following aspects:

- Policies and Regulations: Official documents issued by the National Energy Administration, the National Development and Reform Commission, and local authorities regarding hydropower development, photovoltaic subsidies, grid integration technical standards, and watershed management.
- Technical Standards and Specifications: Documentation from IEEE, IEC, Chinese National Standards (GB), and industry standards covering areas such as turbine efficiency, photovoltaic module performance, power station design, and operational safety.
- Academic Research and Patents: Core journal articles, conference papers, and relevant technical patents sourced from databases like CNKI and Web of Science, focusing on cutting-edge topics such as energy conversion efficiency, new material applications, and intelligent maintenance.
- Project Reports and Industry Analyses: Feasibility studies of power station projects, industry white papers, and market trend forecasts published by authoritative domestic and international consulting institutions.

C. Data Preprocessing

Following the initial compilation, the research team identified several challenges in the collected energy sector data, including diverse and complex formats, dense technical terminology and symbols, inconsistent data units, and variable quality in some scanned documents. Consequently, rigorous and meticulous data preprocessing was essential prior to knowledge base construction. The core objectives of this preprocessing were to clean, standardize, and annotate the data in depth, ensuring it meets the requirements for high-quality knowledge input into large language models. The work primarily addressed three aspects: data format uniformity, content structuring, and semantic integrity.

- Data Parsing and Format Standardization: For original files in various formats such as PDF, Word, HTML, and Excel, information was extracted using direct text extraction for native digital files and Optical Character Recognition (OCR) technology specifically for scanned documents. This ensured complete and accurate digitization of all textual information.
- Content Cleaning and Semantic Coherence Assurance: Noise and non-critical information, such as extraneous headers, footers, page numbers, and advertisement text, were removed from the text. Particular attention was paid

to intelligently identifying and rectifying issues like broken formulas and table misalignments caused by line breaks in technical documents, thereby preserving the semantic continuity of professional terms and complete sentences.

- Domain Knowledge-Enhanced Annotation: Building upon basic cleaning, specialized energy domain lexicons and ontologies were introduced to identify and standardize key entities (e.g., "pumped storage," "PERC cells," "conversion efficiency") and technical parameters within the text. This process lays a solid foundation for subsequent high-precision retrieval and enhanced model comprehension.

D. Retriever Optimization

To enhance the accuracy and relevance of information retrieval from the knowledge base, this study implemented a multi-faceted optimization strategy for the retriever. The methodology integrated dense vector embeddings with traditional keyword-based search, leveraging advanced sentence-transformers to capture deep semantic relationships. A domain-adapted ranking model was subsequently applied to reorder initial results, prioritizing documents based on semantic similarity, contextual alignment, and domain-specific relevance. Specialized techniques were also introduced to handle complex technical queries involving numerical parameters and conceptual relationships. These optimizations collectively ensured that the retriever consistently identified the most pertinent and reliable information fragments to support high-quality response generation.

Knowledge Text Segmentation improve：Given a collection of knowledge texts set $D = \{d_1, d_2, \cdots, d_N\}$, where each document d consists of multiple sentences or paragraphs, a sliding window function $S(d_i)$ is applied to partition the documents into multiple coherent segments.

$$S(d_i) = \{s_{i1}, s_{i2}, \cdots, s_{im}\} \tag{1}$$

Here s_{ij} is the jth segment of document d_i .This article employs fixed w and overlapping step sizes for segmentation, where j denotes the starting position of segmentation and w represents the window length.

$$s_{ij} = d_i[j_w{:}j_w + w] \tag{2}$$

Here j_w denotes the starting position of the segmentation, and w represents the length of the window.

Dynamic Embedding Approach for Word Representation: For each text segment s_{ij} , an embedding model E(x) is employed to map it into a spatial vector:

$$h_{ij} = E(s_{ij}) \tag{3}$$

where $h_{ij} \in R^d$ denotes the embedding representation of s_{ij}, and d represents the embedding dimensionality.

$$h_{ij} = Transformer(s_{ij}) \tag{4}$$

Vector Aggregation via Mean Pooling: The integration of multiple information sources or features is achieved through an aggregation operation on individual word embedding vectors, yielding a more comprehensive and representative representation.

$$h_i = \frac{1}{m}\sum_{j=1}^{m} h_{ij} \tag{5}$$

The representation of a document d_{ij} , which comprises multiple segments s_{ij}, can be derived through the aggregation of their respective vectors.

Cosine-similarity-based search is performed to identify the top-K most similar vectors to the query q from the database $H = \{h_1, h_2, \cdots, h_N\}$.

$$Retrieve(q, H, k) = \underset{h_i \in H}{\mathrm{top}k} sim(q, h_i) \tag{6}$$

And in equation (6), we have use Cosine Similarity Search

$$sim(q, h_i) = \frac{q^T h_i}{\|q\|\|h_i\|} \tag{7}$$

IV. Experiments and Results

A. Experimental setup

The experiments in this study were conducted using Python 3.12. The base model accessed via API was GPT-3.5-turbo, and the text-embedding-3-large model was employed for generating vector embeddings. The experimental hyperparameters are detailed in Table 1.

TABLE I. EXPERIMENTAL SETUP

Parameter name	***Parameter value***	***Description***
Chunk_size	1 200	Character count per document segment
Chunk_overlap	500	Overlapping character count between consecutive document segments
Max_tokens	6 000	Maximum number of semantic units per document
Top_k	5	Maximum number of documents to return
Temperature	1	Temperature of the large language model

B. Experimental Results and Analysis

To comprehensively evaluate the performance of the research model, this study conducted an assessment of the question-answer pairs in the experimental dataset and calculated the average score, as shown in Table 2

As evidenced by the data in Table 2, the proposed model comprehensively outperforms the purely generative baseline (GPT-3.5-turbo) across three key metrics: Answer Similarity, Answer Relevance, and Answer Correctness. This performance gain is primarily attributed to the integrated retrieval-augmented generation (RAG) mechanism.

The model achieved an Answer Similarity score of 0.93, marking an improvement of 0.04 over the base model. This indicates that the retrieval-augmentation mechanism effectively constrains semantic divergence in the generated text. By incorporating relevant external knowledge, the model produces responses that are more aligned with the factual context.

Furthermore, the model attained an Answer Relevance score of 0.79, an increase of 0.04 points. This result validates the synergistic effect of the RAG framework's two-stage architecture: the retrieval module accurately pinpoints critical information, which allows the generation module to concentrate its output on the retrieved content, thereby reducing redundancies and mitigating tangential responses.

Most notably, the model scored 0.63 in Answer Correctness, which constitutes a significant improvement of 0.05 points. This demonstrates that the external knowledge base substantiates the factual reliability of the generated content. The retrieved authoritative data serves as a factual anchor for the generation process, significantly curbing the phenomenon of hallucination.

TABLE II. EVALUATION AND COMPARISON OF RESULTS

item	*ours*	*GPT-3.5-turbo*
Answer Similarity	0.93	0.89
Answer Relevance	0.79	0.77
Answer Correctness	0.63	0.58

C. Analysis of question-answering results

1. Professional Question Answering

In the domain of professional question answering, our model demonstrates a marked superiority over the base model, a benefit derived from our optimized local knowledge base and retrieval processes. While the base model explains terminology based on general knowledge, our model delivers responses that are strongly correlated with specialized expertise.

Question:

What are the main technical indicators for the recommended cascade development plan of the Kelong River Basin hydropower project?

GPT-3.5-turbo:

Certainly, regarding the cascade development plan for the Kelong River, this is a highly specialized issue in hydropower planning. It is important to clarify that the "Kelong River" is not a widely recognized river with publicly available detailed development plans.

Therefore, I cannot provide an actual, specific cascade scheme for the Kelong River or its precise technical indicators. However, I can construct a typical and representative template of a cascade hydropower development plan for a mountainous river.

Ours:

For information related to the Kelong River Basin, please refer to **Table 4.1.1-3 in the XX Report**, which contains the main technical indicators of the recommended cascade development plan for hydropower exploitation in the Kelong River Basin.

D. further work

Optimized large language models (LLMs) have demonstrated notable advantages in energy-related professional Q&A; however, the following limitations persist: Firstly, while the responses can currently retrieve images and text from the local knowledge base, there remains a significant deficiency in associating tabular data and mathematical formulas. Secondly, the issue of logical contamination persists: even when the model provides correct answers based on the local knowledge base, it may still fabricate or generate erroneous responses if users strongly contradict it. These issues not only compromise the accuracy and credibility of the model's outputs but also risk misleading users, thereby undermining the reliability and effectiveness of the model in practical applications.

V. CONCLUSION

Based on the experimental results, this study validates the effectiveness of Retrieval-Augmented Generation (RAG) in vertical domain question-answering within the energy sector. The advantages are particularly pronounced in terms of content relevance. However, there remains room for improvement in answer correctness. By constructing a large language model for energy domain QA based on RAG technology, this work effectively addresses the issues of knowledge hallucination and inaccuracy inherent in traditional base models when applied to specialized domains. The experimental results demonstrate the synergistic effect between the construction of a vertical energy knowledge base and the RAG retrieval-generation mechanism. The application of RAG technology in professional fields can significantly enhance semantic understanding of vertical domains, reduce ineffective responses, and mitigate factual errors.

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